Binder Generation with BindCraft

Table of Contents

1. Prerequisites

1.1 Install VS Code

Download and install VS Code from the official download page:

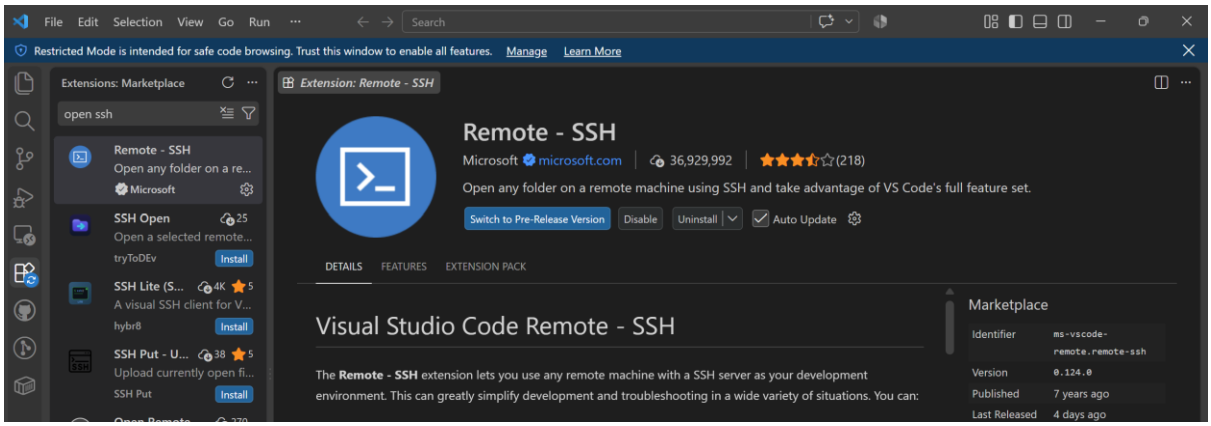
https://code.visualstudio.com/download?__exp__download=fb315fc982

1.2 Github Account

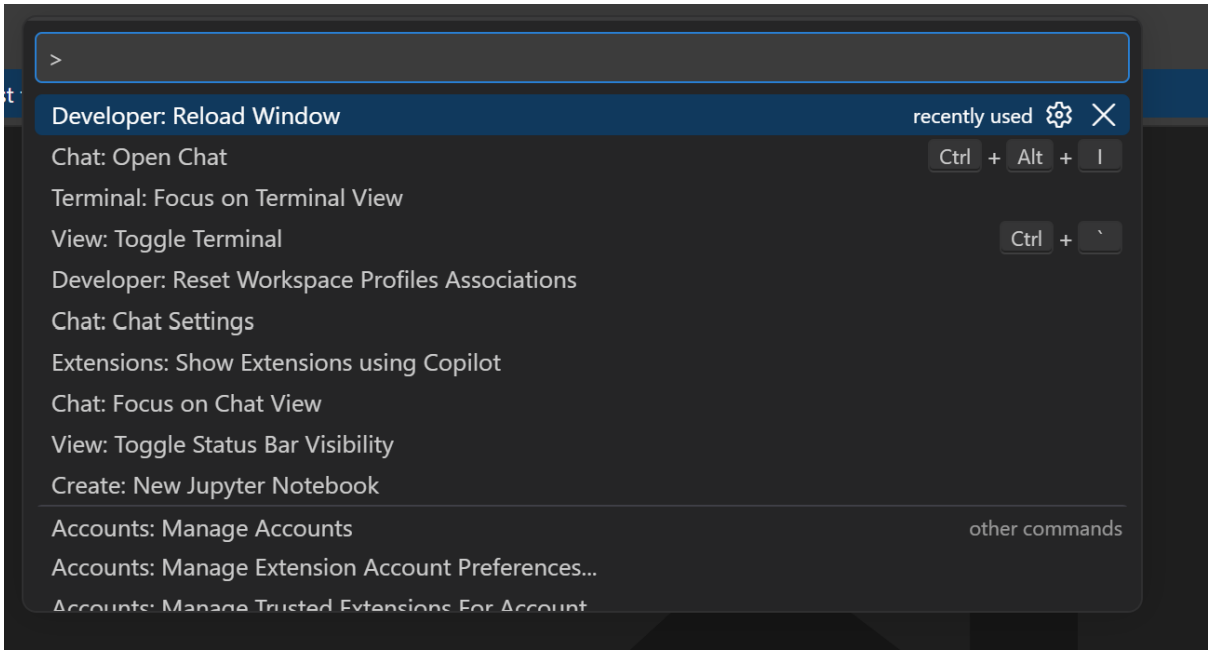
Create a GitHub account at https://github.com/ if you do not already have one.

1.3 Remote- SSH extension

Install the **Remote – SSH** extension in VS Code. This extension enables you to connect to an Ewok machine from your local computer and access its computational resources directly through VS Code.



After installation, open the **Command Palette** and run **Developer: Reload Window** to ensure that the extension loads correctly.

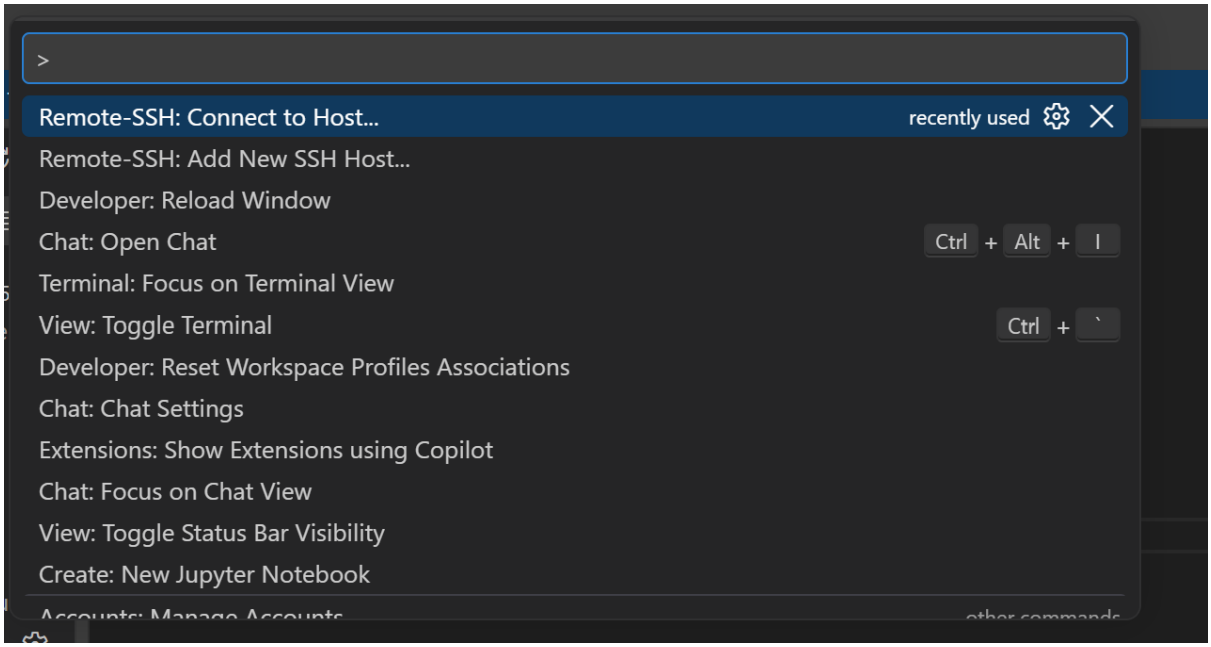


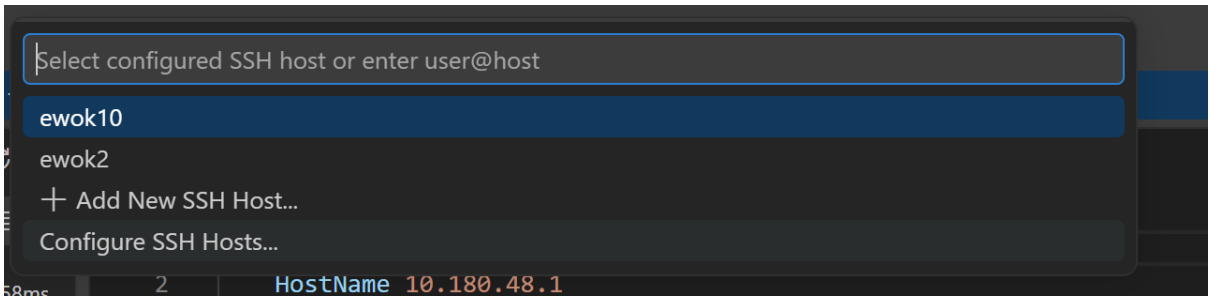
1.4 BindCraft Installation on Ewok

Before proceeding, ensure that BindCraft is installed and configured on your Ewok machine. Refer to the *BindCraft Ewok Setup Tutorial* for installation instructions.

2. Connect to the Ewok Machine

In VS Code, open the **Command Palette** and select **Remote-SSH: Connect to Host…**, followed by **Configure SSH Hosts…**.





Add an entry for your Ewok machine using the following configuration format:

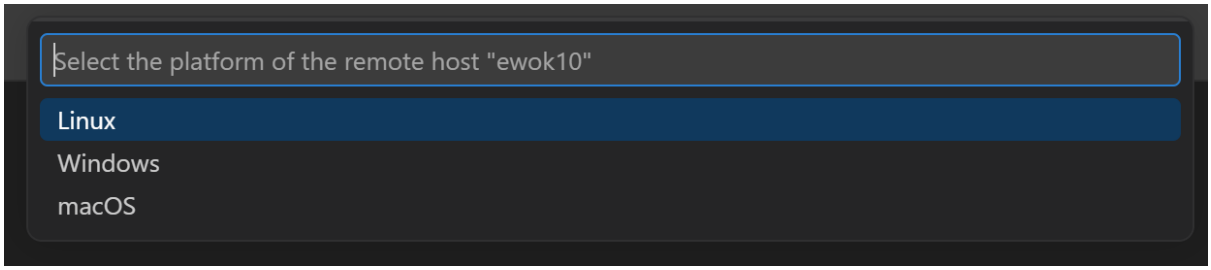
```
Host <ewok_machine_number>
  HostName <your_ewok_machine_ip_address>
  User <your_ewok_username>
  Port <your_ewok_port>
```

Example configuration:

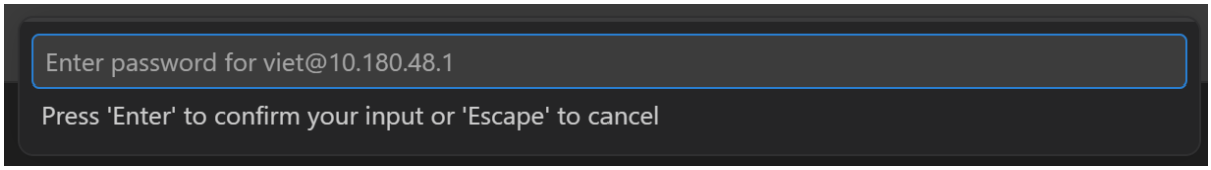
```
Host ewok2
  HostName 11.181.47.0
  User viet
  Port 11111
```

Note: Contact Zhen He or Aaron Haris for setting up and access your Ewok account.

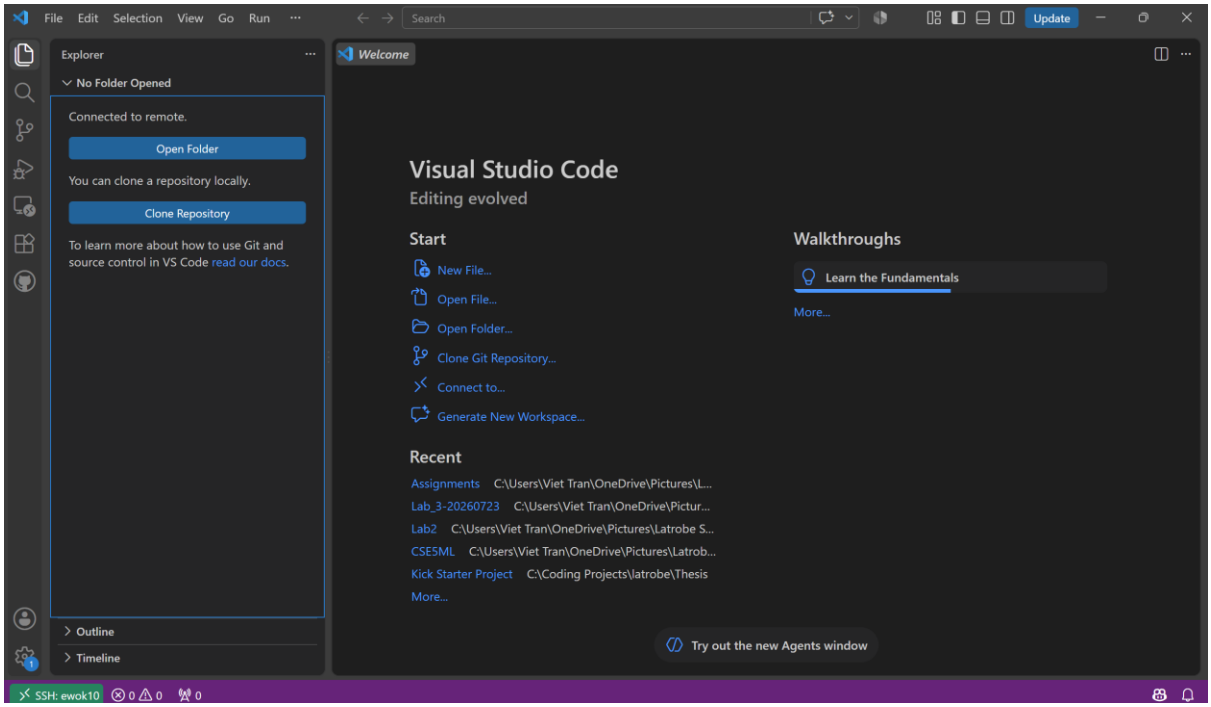
If VS Code prompts you to select the remote host's operating system, choose **Linux**



Enter your Ewok password to complete the connection.

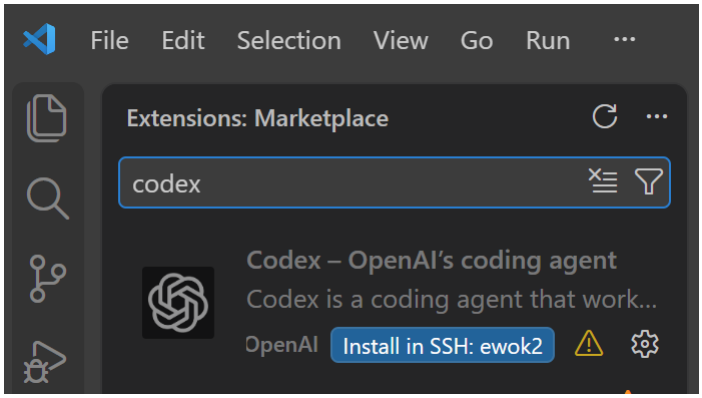


Once the connection is established, the Ewok host name should appear in the bottom-left corner of the VS Code window.



Optional: Install Codex

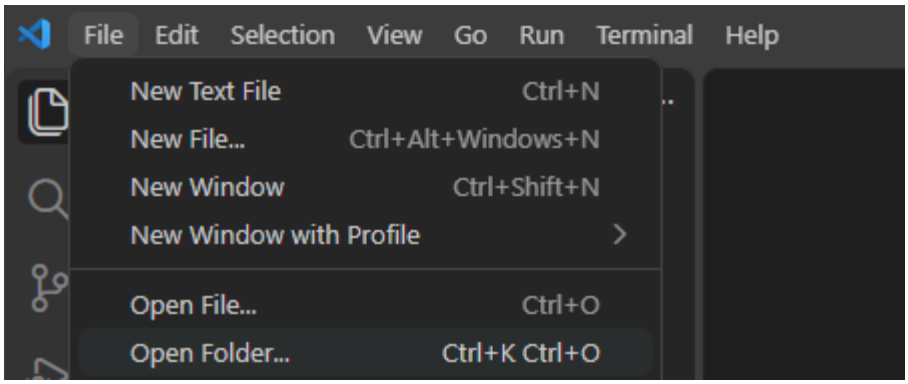
If you have a ChatGPT account, you can optionally install the Codex extension on the Ewok host for AI-assisted work within the project.



3. Binders Generation with BindCraft

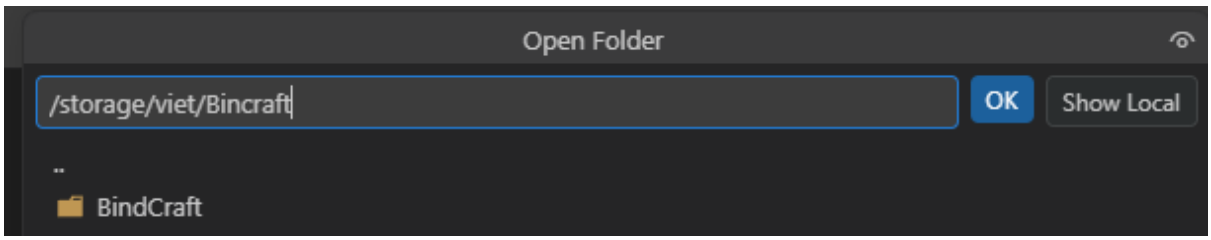
3.1 Open the BindCraft Project

After connecting to the Ewok machine, select **File → Open Folder** in VS Code.



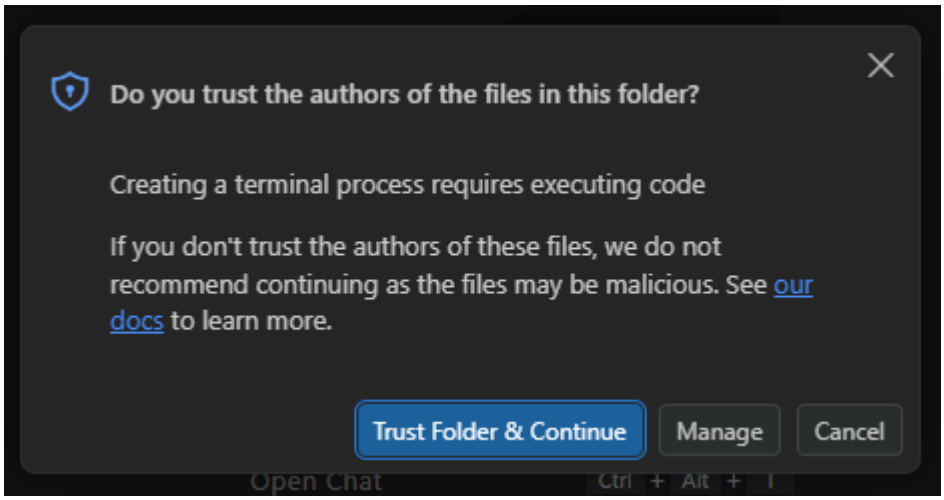
Open the directory in which BindCraft is installed. Typical installation locations include:

- /home/<username>/BindCraft
- /storage/<username>/BindCraft



Replace <username> with your Ewok username.

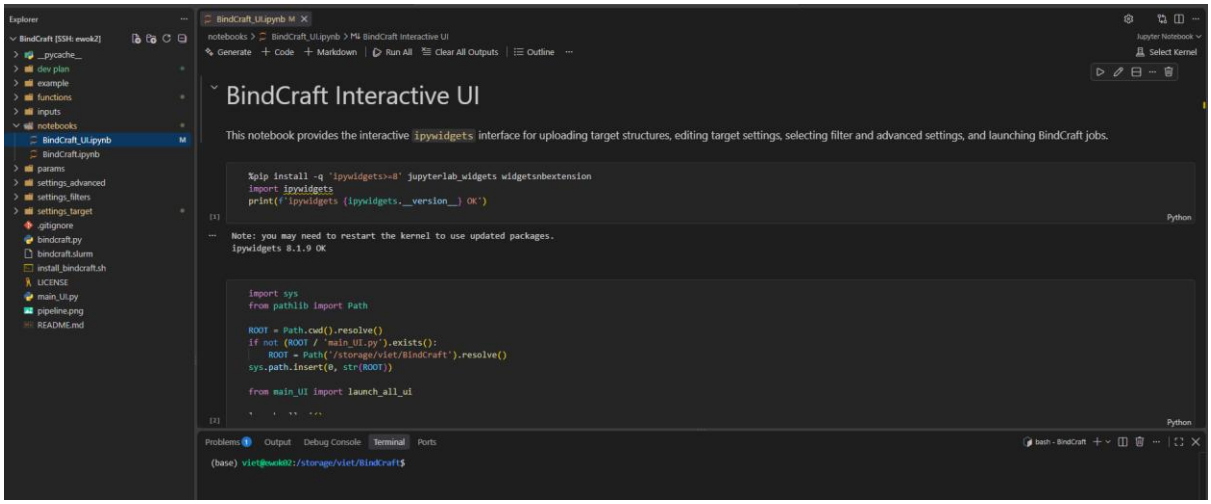
If prompted to confirm whether you trust the folder, select **Trust Folder & Continue**.



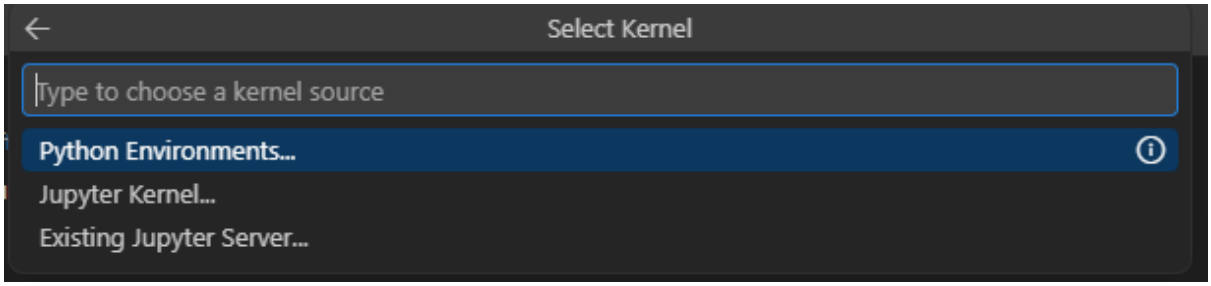
3.2 BindCraft UI

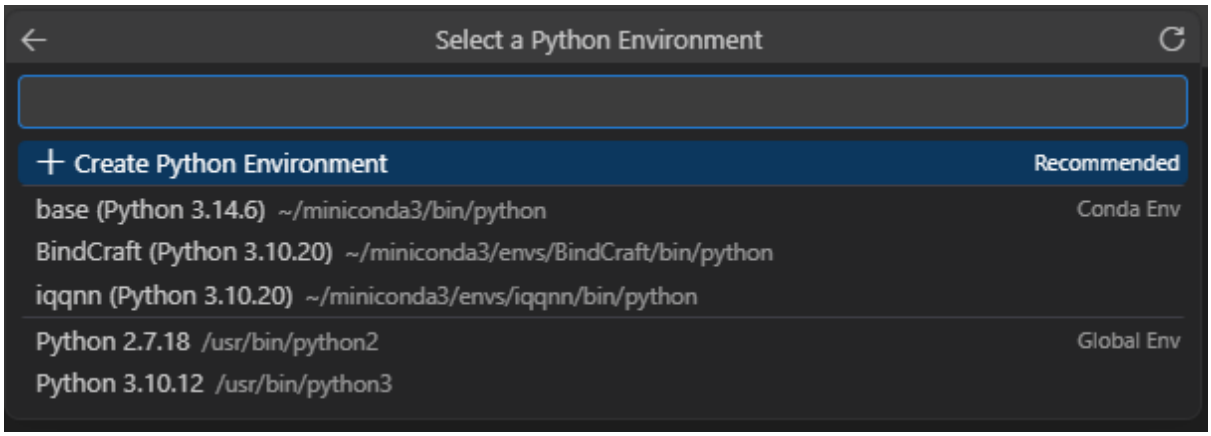
3.2.1 Set Up the Notebook

In the **Explorer** panel on the left side of VS Code, open notebooks/BindCraft_UI.ipynb

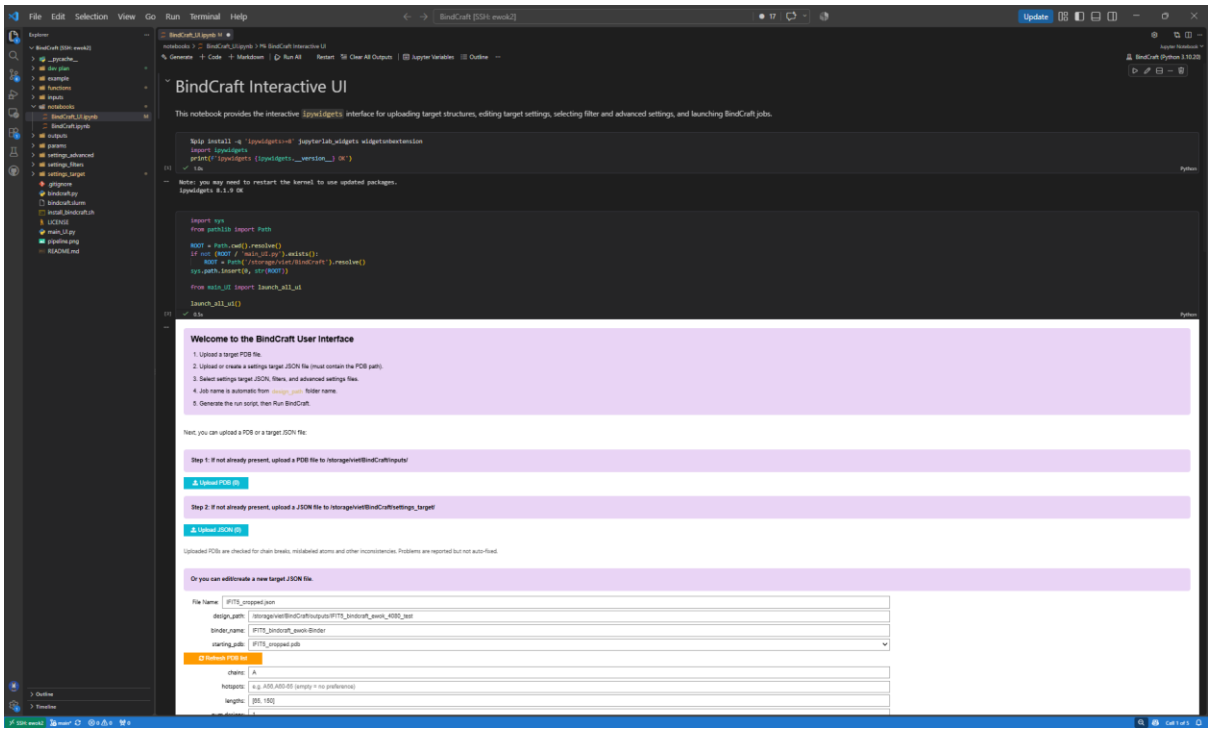


Click **Select Kernel**, choose **Python Environments**, and select the **BindCraft** environment.



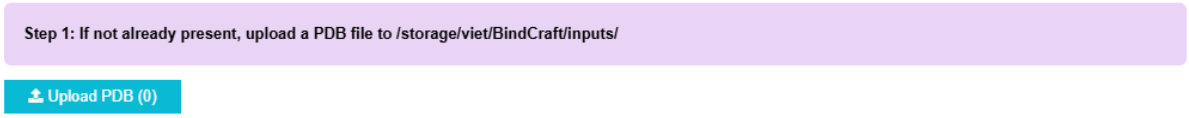


After kernel successfully connected, click **Run All** to execute the notebook cells, install the required user interface packages, and display the interactive BindCraft interface.

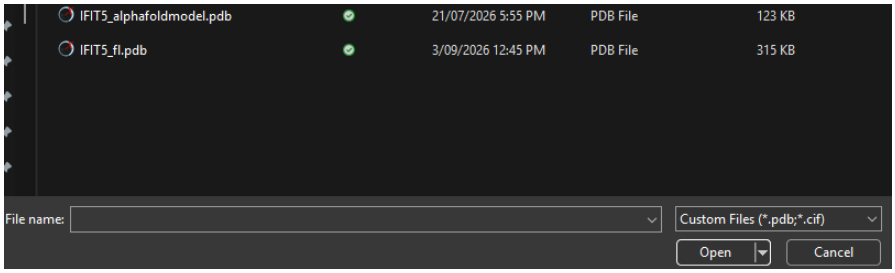


3.2.2 Run a bindCraft job via the UI

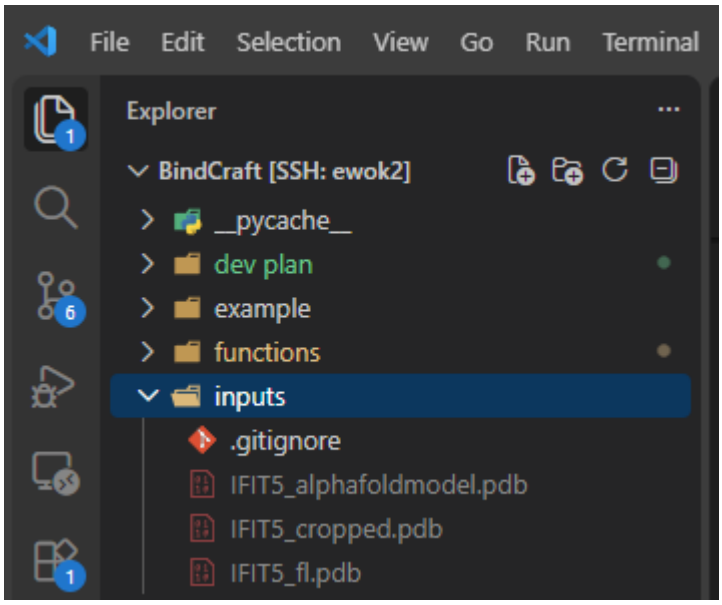
Step 1: Upload a Target PDB file



Use the **Upload PDB** control to select and upload the target PDB file from your local computer.



The uploaded file will be saved in the inputs/ directory within the BindCraft project:

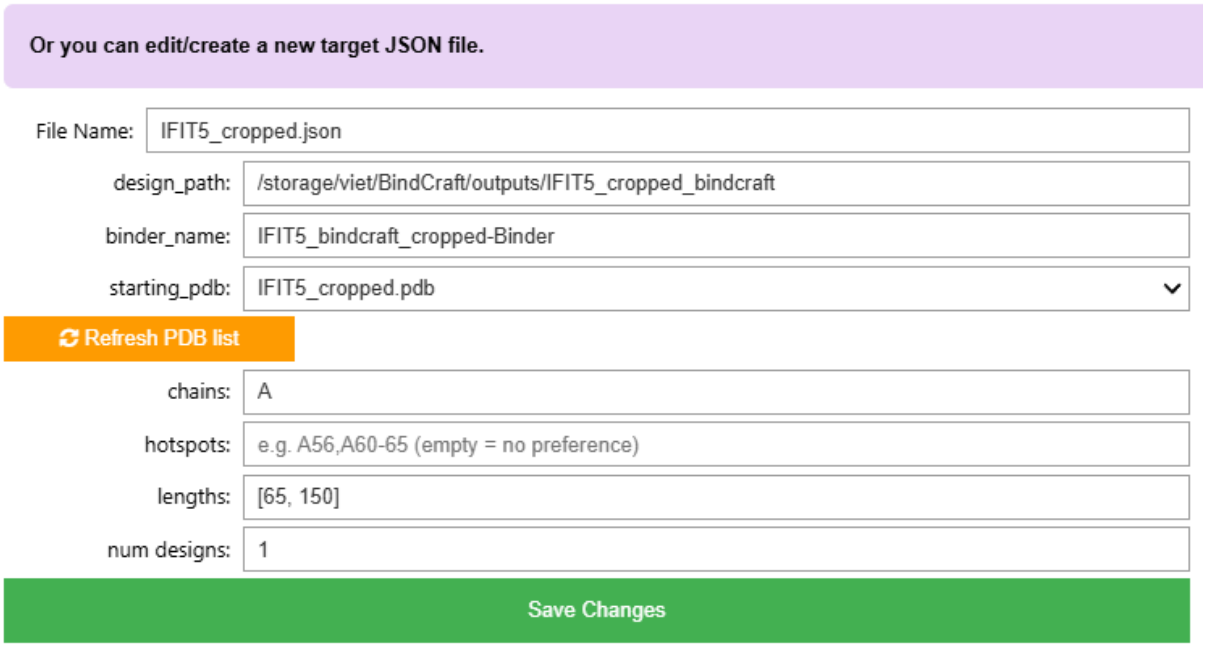


Step 2: Configure the Target Settings

The target settings JSON file specifies the target structure against which BindCraft will generate binders. It also defines the binder naming prefix, output directory, target chains, hotspot residues, binder length range, and requested number of accepted designs.

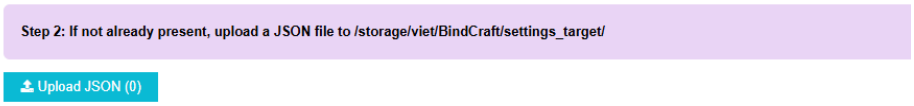
If you do not already have a target settings JSON file, create one directly through the user interface by completing the following fields:

Field	Description
File Name	Target settings JSON filename.
design_path	Directory for outputs of a BindCraft run.
binder_name	The prefix used for generated binder filenames, for example, IFIT5_bindcraft_ewok-Binder.
starting_pdb	The uploaded target PDB file.
chains	The target chains to include, for example, A or A,C. Other chains will be ignored.
hotspots	The target residues intended for binder interaction, for example, A56,A60-65. Leave this field empty for automatic selection.
lengths	The minimum and maximum binder lengths, for example, [65,150].
Number designs	The requested number of accepted binder designs.

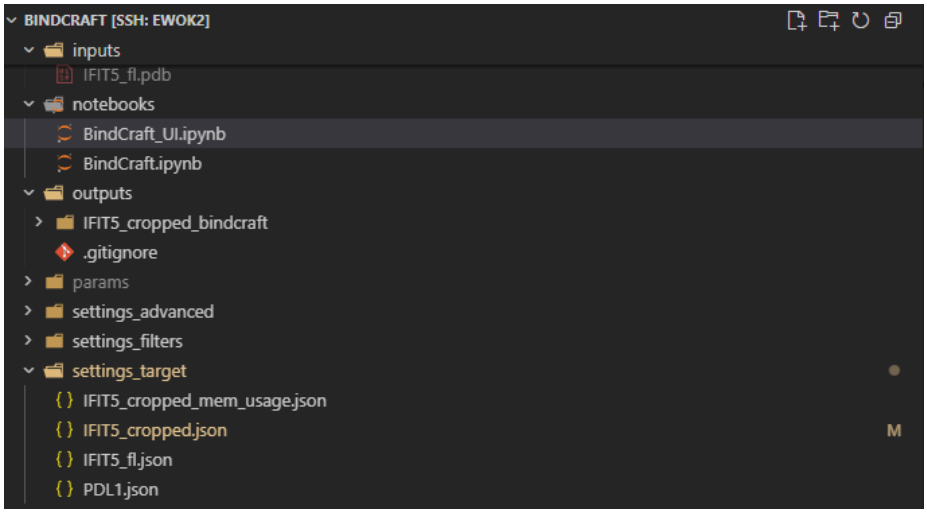


After completing the fields, click **Save Changes**.

Alternatively, upload an existing target settings JSON file that corresponds to the uploaded target PDB, rather than creating a new configuration manually.



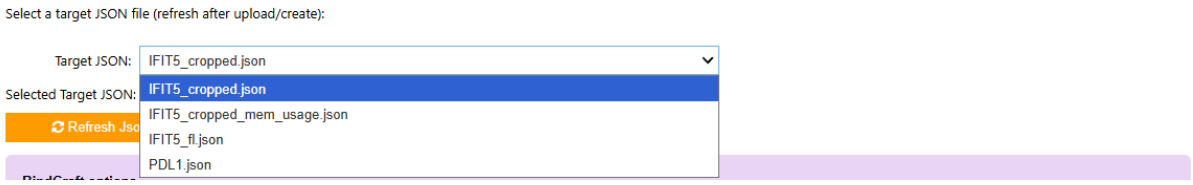
Both uploaded and newly created target settings files are saved in the project's settings_target/ directory.



Step 3-6: Run BindCraft on specific target setting

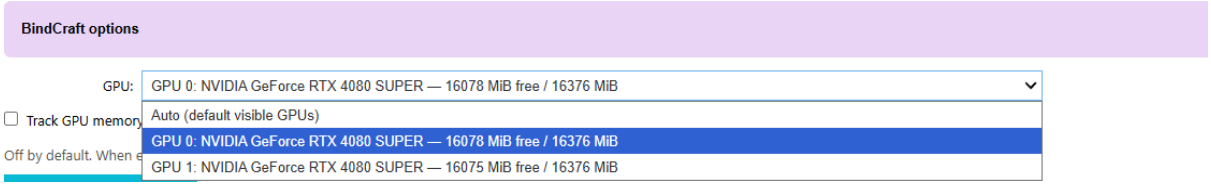
Target setting selection

From the Target JSON dropdown menu, select the target settings file that you created or uploaded.

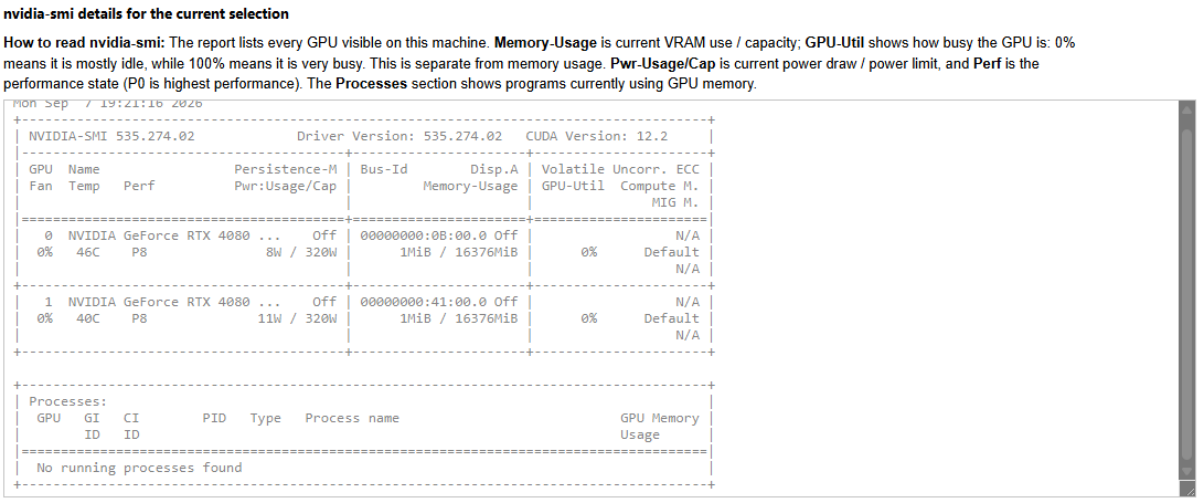


Configure the GPU

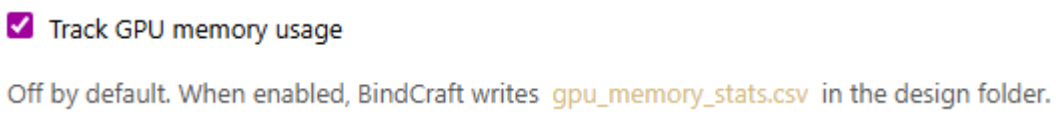
Select the GPU to use for the BindCraft job.



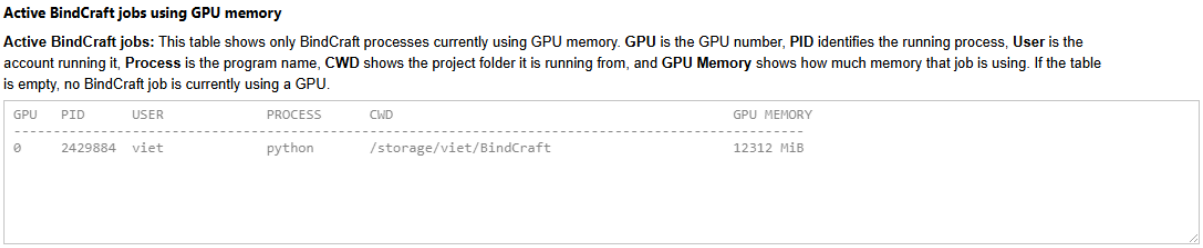
Review the **nvidia-smi** panel for information about the available GPUs and their current usage.



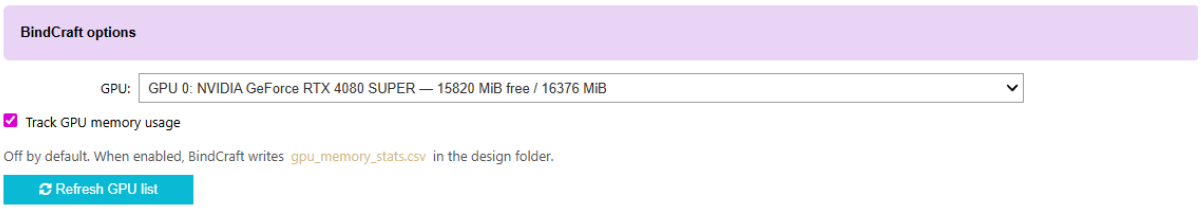
Binder generation against large target proteins can require substantial GPU memory. Enable **Track GPU memory usage** to monitor memory consumption and assess it against the selected GPU's capacity.



The **Active BindCraft jobs using GPU memory** panel displays BindCraft jobs currently using GPU resources.



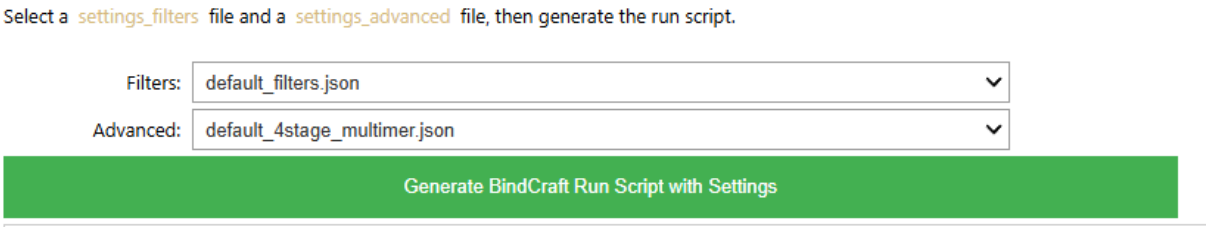
Click **Refresh GPU list** to update the GPU information and active-job displays.



Configure binder filtering and design validation

The **Filters** setting defines the quality criteria that binder designs must satisfy to be accepted. These criteria include confidence scores, interface quality, structural clashes, and binding energy.

The **Advanced** setting controls how BindCraft generates and validates binders, including the design algorithm and the AlphaFold2 and ProteinMPNN settings.



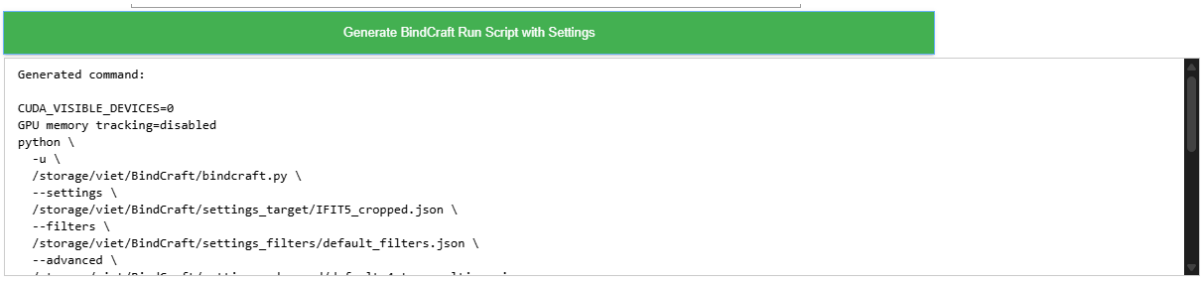
For the default configuration, select default_filters.json under **Filters** and default_4stage_multimer.json under **Advanced**.

For further information, refer to the following resources:

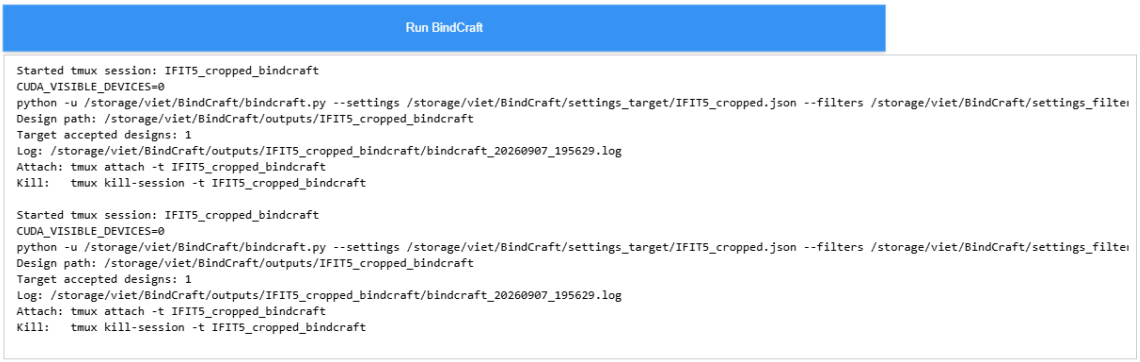
- **Filter settings:** https://github.com/martinpacesa/BindCraft#filters
- **Advanced settings:** https://github.com/martinpacesa/BindCraft#advanced-settings
- **BindCraft paper:** https://doi.org/10.1101/2024.09.30.615802

Generate the run script and launch the job

Once all settings have been configured, click **Generate BindCraft Run Script with Settings**.

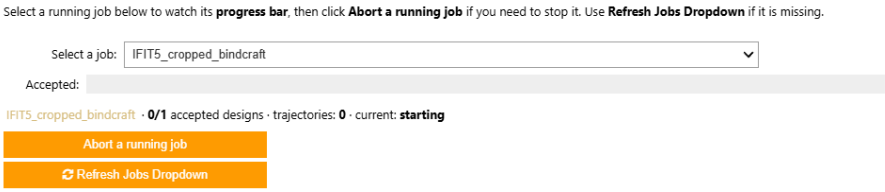


Then click **Run BindCraft** to launch the job using the generated script.

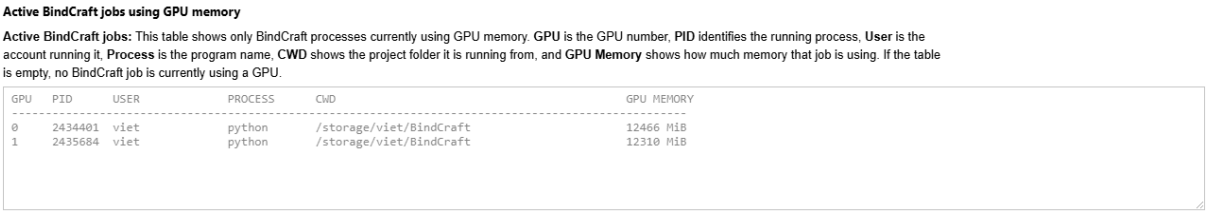


Monitor and manage jobs

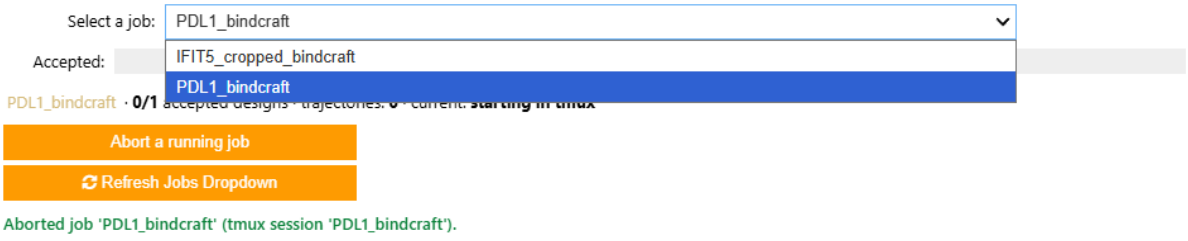
Monitor the selected job's progress in the status section at the bottom of the user interface.



Multiple BindCraft jobs can run concurrently. Assign a different **design_path** to each job so that their outputs are stored in separate directories.



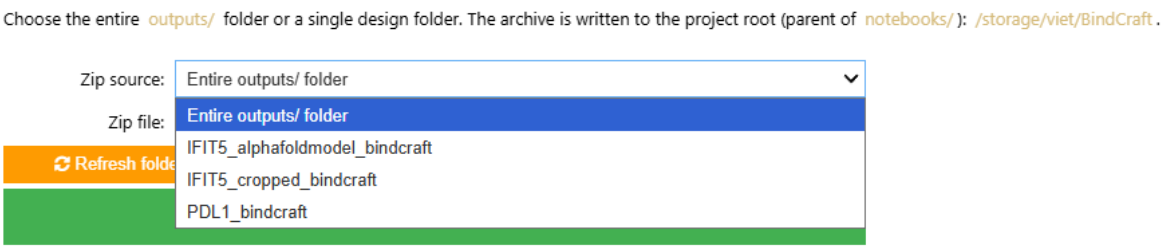
To stop a job, select it from the job dropdown menu and click **Abort a running job**. If a running job is not listed, click **Refresh Jobs Dropdown** to update the available selections.



Step 7: Compress and Download the Outputs

BindCraft output files can be large. Compress the required outputs into a ZIP archive before downloading them.

In the **Zip source** dropdown menu, select either the entire outputs/ directory or an individual design output folder, then create the archive.



The ZIP archive will be saved in the BindCraft project's root directory. To download it, locate the .zip file in the VS Code **Explorer** panel, right-click it, and select **Download…**. Choose a destination on your local computer and save the file.